



The company you keep: Syndicate formation in venture capitalists' cross-border investments[☆]

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ABSTRACT

This paper studies how venture capitalists trade-off agency and resource considerations when syndicating for cross-border investments. We argue that syndicate size and the decision to partner with local VCs involves a complex trade-off between the potential agency costs from multiple VCs in one syndicate and the benefits of foreign VCs with complementary resource endowments. Based on data of 15,725 cross-border VC investments we show that, in the presence of geographic and psychic distances between VC syndicate members, VCs reduce complexity from syndicate size (agency perspective) but are more likely to invite local VC co-investors (resource perspective). VC firms with high international recognition rely less on other syndicate members' resources and minimize agency costs through smaller syndicates. VC firms with high host-country recognition face lower local partner search costs and have a higher probability to have local partners in the syndicate. Our paper contributes a framework for syndication in cross-border VC investment that recognizes the importance of distance between VCs and the role of recognition of VCs as a valuable resource in the syndication trade-off.

1. Introduction

Venture capital (VC) has played a fundamental role in financing innovative companies (Young, 2003), offering both financial and strategic value to their investees (Chahine et al., 2019; Park and LiPuma, 2020). Indirectly, Venture Capitalists (VCs) play an essential role in the international proliferation and professionalization of corporate governance (Lerner & Nanda, 2020). As such, venture capitalists combine roles of international finance, governance, and strategy in a uniquely relevant way (Amit et al., 1998).

Because of the difficulties involved in applying their hands-on business model across distance, VCs were long characterized by geographically focused business strategies (e.g., Guler & Guillen, 2010; Sorenson and Stuart, 2001; Sorenson and Stuart, 2008; Stuart and Sorenson, 2003a; Stuart and Sorenson, 2003b; Cumming & Dai, 2010). More recently, VC firms have increasingly internationalized their investments, defying capital market liabilities of foreignness (LOF) (Gu et al., 2019; Zaheer, 1995). In 2020, 30 % of VC investments were cross-border investments compared to 20 % in 2000 (see Appendix 1, Prequin, 2021).

This increase in cross-border investment activity is reflected in

growing academic interest. Cross-border VC investments are more prone to LOF than other capital markets (Bell et al., 2012; Gu et al., 2019) because the VC business model builds on access to lucrative investments (Lu & Hwang, 2010) and hands-on development of target companies (Buchner et al., 2018) both of which require local know-how, information and social capital. Despite substantial CMLOF, cross-border VC investments have shown to outperform purely local VC investments, potentially due to complementary resources from foreign VCs (Chemmanur et al., 2016; Cumming et al., 2016). Yet, it is unclear which resources or strategies allow these internationalizing VC firms to overcome LOF and outperform domestic investments.

One particularly important strategy in international VC investment is syndication. In VC syndication, “form follows function” (Pichler & Wilhelm, 2001, p.2237). Syndication theory, herein, predicts an intricate trade-off between the costs and benefits of collaboration (Filatotchev et al., 2006). On the one hand, co-investment allows VCs to pool general and host-country-specific resources (Chemmanur et al., 2016; Dorobantu et al., 2020). On the other hand, collaborating across cultural distances has been shown to impede VC collaboration (Dai & Nahata, 2016). VCs and entrepreneurs thus rely on an optimal trade-off between resources and agency costs in the syndicate to create innovation,

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growth, and, ultimately, financial returns. Efficient VC syndicates can pool resources essential to the development of the investee company, ultimately enabling growth, innovation, and further investment (Hochberg et al., 2007; Hochberg et al., 2010). Dysfunctional syndicates can trigger moral hazard and resource overlap.

Understanding VC syndication is vital to understanding the internationalization of this particularly important form of financing. VC investors and entrepreneurs rely on an optimal trade-off between resources and agency costs to create innovation, growth, and financial returns. Similarly, policymakers should understand the costs and benefits of VC syndication to create institutions that facilitate cross-border VC investment in their countries.

The composition of syndicates (and its consequences) has been studied in other disciplines (Chemmanur et al., 2016; Guler and Guillén, 2010) and empirical contexts ranging from R&D alliances (Dalziel, Gentry & Bowerman, 2011) to board interlocks (Zona, Gomez-Mejia & Withers, 2018). In investment syndication, prior research (a) has commonly applied a crude theoretical distinction between foreign and local investors, (b) has not acknowledged the fact that some VCs have local or global recognition and status, and (c) applied the concept of cultural distance rather than the conceptually more related concept of psychic distance.

Most relatedly, in IB, Dai & Nahata (2016) study how cultural distances affect syndicate size and the participation of local VCs. However, their analysis focuses on the role of distance between the VCs and the target investment country of a local VC, depending on the experience of the involved foreign VCs. Our study focuses on how distances between collaborating partners (i.e. syndicate members) alter the trade-off between agency cost and pooled resources (Dorobantu et al., 2020; Gianetti and Yafeh, 2012; Müllner and Dorobantu, 2022). In addition, we hypothesize that general and host-country-specific recognition of a VC fund offers privileged access to ad-hoc network resources and reduces the need to pool resources through broad syndication. We argue that recognition differs from experience conceptually and show some empirical evidence to support this claim.

To empirically test our predictions, we leverage a uniquely comprehensive and recent dataset from Preqin, which comprises 15,725 syndicated VC cross-border investments between 2010 and 2020. This multi-country sample includes investments from 67 home markets to 55 host markets. We combine this data with inter-partner psychic distance measures (Dow, 2021). Therefore, we deviate from existing literature that focuses on cultural distance. Further, we leverage unique data on VCs' host-country recognition derived from Google Trends to capture VC status (Guler & Guillén, 2010). This recognition-based variable is more capable of capturing not only a VC's internal resources from past investment history but, more accurately, a VC's status, recognition, and social capital in the eyes of critical partners in this network-based business model (Guler and Guillen, 2010; Guler and Guillén, 2010; Mingo et al., 2018).

We find that geographic and psychic distance (Dow & Larimo, 2009) are important determinants of VC syndicate formation. Distances between syndicate members appear to create costs of agency and lead to smaller syndicates with more local VCs. Further, our paper finds compelling evidence that foreign VCs' general and host-country-specific recognition affects how they syndicate. Specifically, we find that general recognition results in significantly smaller syndicates indicating that VCs with strong global recognition do not need to rely to an equal extent on resource pooling through syndication. This may be the case because their global recognition allows them to deploy further resources globally in case needed within the local investment. Surprisingly, we further find that foreign VCs' local recognition in a host market is positively associated with the participation of a local partner in a syndicate. This indicates that VCs with strong local recognition may have better access to local partners and appear to value the resource contributions of local VC partners more than the potential of increasing agency costs. One possible explanation for this finding may be that foreign VCs with strong

host-country recognition might benefit from a local recognition bonus that reduces the costs of local partner search (specifically as compared to VCs with strong global recognition, only).

This study contributes a theoretical framework of the resource-agency cost trade-off in VC syndication to the growing literature on cross-border VC in international business. More specifically, it highlights the importance of psychic distances between co-investing VCs and of limiting the size of syndicates due to agency concerns. Second, it shows the specific relevance of recognition for syndication. It shows that VC general recognition is a valuable resource to reduce the need for larger syndicates and local partners. However, VC firms with high recognition in the local market are more likely to co-invest with local VCs, potentially as they can capitalize on their recognition in the two-sided matching process of syndication (Guler & Guillén, 2010).

2. Theory and hypotheses

2.1. Cross-border venture capital investment and research

In the 2000s, cross-border VC emerged as a multidisciplinary research domain with publications spanning finance, economics, strategy, and entrepreneurship as well as international business and economic geography. (Devigne et al., 2018; Wright et al., 2005). As a hands-on business model, cross-border VC investment suffers intensively from liability of foreignness (LOF).

VCs face higher agency costs due to information, monitoring, and access disadvantages when investing across borders compared to home investments. Nevertheless, investments involving foreign VCs (i.e., cross-border investments) have been shown to outperform investments with purely local investors (Cumming, Knill, & Syvrud, 2016). This is likely because foreign investors can provide the investee with valuable resources, as defined by the resource-based view of the firm (RBV). It is, however, unclear what exactly creates agency costs and which resources provide value in cross-border investments.

Literature has also provided results that indicate that the trade-off between agency and resources may be contingent on the economic integration of the countries involved (Alhorri et al., 2008) or distance between the investor and the target country (e.g., institutional-, geographic-, cultural distance) (Moore et al., 2015). Institutional distances elevate information asymmetries and agency problems on firm resources (e.g., knowledge, network, status) and on the quality of local institutions supporting VC investment (e.g., liquid capital markets, governance) (e.g., Alvarez-Garrido & Guler, 2018; Bertoni & Groh, 2014; Chahine & Sade 2019; Dai, et al., 2012; De Prijcker et al., 2012; Hain, et al., 2016; Madhavan & Iriyama 2009; Moore, et al., 2015; Schertler & Tykvova, 2011). In this study, we follow this line of reasoning but do not focus on the institutional setting of the investor-investee dyad as in previous research, but on distances between syndicated VCs.

2.2. Syndication in cross-border VC

In practice, more than half of all VC is invested through syndicates (Lutz & Nörthemann, 2020). The formation of syndicates is a complex multi-stage matching process. During the process, a lead VC commonly helps approach potential partners. Herein, the lead VC's status serves as an essential signal of quality (Guler & Guillen, 2010). Further, prior co-investment and trust have also been shown to play important roles (Bottazzi et al., 2016). Adding to the complexity, VC syndication, in practice, involves a two-sided matching process rather than a one-sided decision by VCs (Fu et al., 2019; Mindruta et al., 2016). In other words, both investment and co-investment decisions are made on both sides. They are not, as often portrayed (or empirically modeled), unilateral decisions of VCs imposed on targets and co-investors. Instead, VC (co-) investment is an environment of closed networks with reciprocal relationships that rely on signaling to external stakeholders (Sorenson and

Table 1
Literature overview.

Paper	Purpose	Variables of interest	Key findings
1. Foundations			
Lerner (1994)	VC syndicate joining round timing	Valuation change, VC age, VC size	Established VCs typically syndicate in 1st round, might also join later when IPO likely
Lockett & Wright (2001)	Syndication motives	Finance, resource-based, deal flow	Motive depends on VC type: finance is important, but RBV-focus in early-stage VC
Sorenson & Stuart (2001)	Spatial distribution in VC	Network position, geographic distance	VC networks from VC investing help diffuse information and expand spatial radius
Brander, et al. (2002)	Syndication Motives	Narrowly & broadly defined syndication	Syndication more likely for adding value to projects than for purpose of 2nd opinion
Cumming (2006)	Syndication and adverse selection	Firm types/ syndication, finance types	Financing of low-quality firms is less problematic in syndicated deals due to screening
Manigart, et al. (2006)	Syndication motives	Finance, deal flow, selection, value-add	Finance is the most important motive: portfolio mgmt. strategy drives syndication
Hochberg et al. (2007)	Effect of VC networks on performance	Network characteristics	Better connected VCs generate better performance for themselves and their investee companies
Cumming & Dai. (2010)	Local bias in VC	Network distance, local bias	Geogr. diversified networks facilitate e. g., info flow/ monitoring and reduce local bias
Hopp & Rieder (2011)	Syndication & industry focus	Syndication, VC industry concentration	Syndication associated with industry focus, highlighting managerial resource motives
Hochberg et al. (2015)	Network formation in VC	Similarity vs. novelty of ties	No evidence for similarity bias in new-tie development, suggesting that agency conflicts in partnering are dominated by the desire to accumulate higher levels of certain resources.
Bottazzi, et al. (2016)	Effect of trust in VC	Trust between countries, syndication	Low levels of trust between citizens of two countries render syndication more likely
Bayar, et al. (2020)	Theoretical model	Industry complexity, VC specialization	VCs syndicate in more complex industries and when VCs are more specialized
2. Syndicate formation in international VC and syndicate characteristics			
Dai, et al. (2012)	Analyze local partner & exit perf.	Joint inv., stage, exit likelihood	Foreign/local synd. more likely to invest at early stage & are more likely to exit
Tykova & Schertler (2014)	Local partner & geo distance	Distance, cross-border inv., synd.	Syndicating w/ local partners helps overcome drawbacks

Table 1 (continued)

Paper	Purpose	Variables of interest	Key findings
Huang, et al. (2015)	Syndication in China	Stage, PFC age, local office, local VC	of geo (but not inst.) distance Foreign VCs co-invest w/ local VCs in later rounds/mature PFCs, office = substitute
Chemmanur, et al. (2016)	VC complementarity	Foreign/local-VC-synd., exit likelihood	Complimentary skills of foreign/local VCs associated with higher exit likelihood
Cumming et al. (2016)	Optimal syndicate structure	Number and capital share of international VCs, countries	International VC syndicates create superior firm-level investee performance relative to more domestic syndicates.
Dai & Nahata (2016)	Culture and syndication	Cultural distance, syndication, local VC	Cultural disparity between VC/PFC leads to smaller syndicates (fewer local partners)
Hain, et al. (2016)	Local partner & geo/cultural distance	Distance, cross-border VC flows, synd.	Syndicating w/ local VC mitigates negative effects of geo/cultural distance
Khavul & Deeds (2016)	Syndication in Israel	Inv. experience, status, local partner	Combined co-investor experience & high status foster foreign-local VC syndication
Liu & Maula (2016)	Foreign/local partnering	Country/PFC uncertainty, local partner	PFC- (country-) uncertainty increases (decreases) likelihood local partnering
Meuleman, et al. (2017)	Foreign partner selection	Prior ties, institutions, foreign partner	Direct ties drive future partnering; indirect ties/ institutions decrease need for direct ties
Zhang, et al. (2017)	Pior ties & syndication	Tie density, synd., syndicate size	Tie density promotes future syndication, more important as syndication size grows
Falconieri, et al. (2019)	Syndicate size & IPO perf.	Synd. size/ diversity, IPO underpricing	Greater syndicate size and diversity associated w/ higher IPO underpricing
Pezeshkan, et al. (2019)	Syndication in China	Risk, knowledge, local VC	Knowledge capabilities render foreign VCs attractive, increase local partner likelihood
Kim & Park (2021)	Synd. size & performance	Synd. size, geo dispersion, reput., perf.	Inverted-U-relationship of synd. size & performance, increasing coordination costs
Qiu, et al. (2021)	Syndication in China	Cult. dist., reputation, synd., synd. size	Cult. distance btw. home/host market impacts syndication likelihood and synd. size

Literature overview of selected research contributions on syndication and exemplary listing of most relevant, syndication-related content, cult. dist. = cultural distance, inst. dist. = institutional distance, invest. = investments, perf. = performance, PFC = portfolio company, reput. = reputation, synd. = syndication.

Stuart, 2001; Sorenson and Stuart, 2008).

A VC syndicate typically encompasses a lead investor and at least one non-lead VC (Wright & Lockett, 2003). In a domestic syndication context, Hochberg et al. (2007) indicate that a particular VC's network

position helps source high-quality deals and add value to portfolio companies, thus being linked to better investment performance both on a fund and investee level. Certain features of a VC's network position exhibit an essential performance link. This includes a VC's network size, incoming syndicate invitations, and access to well-networked peers.

Other studies take one step back from a syndication-related performance analysis and put the spotlight on how syndicates are composed. Hochberg et al. (2015) examined organizational tie formation among VCs from a similarity perspective since disparate resource endowments might aggravate agency conflicts. The authors indicate that similarity motives do not mainly drive VCs. VCs instead engage in resource aggregation (i.e., cumulative advantage via combining more resources within a specific category such as capital) and resource distribution (i.e., trading different kinds of resources). Specifically, VCs seek a) cumulative advantage regarding network and investment scope (i.e., diversity of prior investment activities) and dissimilarity in experience, as well as b) resource trades between capital and value-added resources (e.g., experience, network, investment scope).

While syndication provides relevant opportunities for VCs, it introduces additional agency costs. Agency challenges increase, for example, with growing syndicate size or increasing competition between VCs (Jääskeläinen, 2012). VC firms use formal agreements to manage the horizontal agency relationships with their co-investing VCs, but there are limits to contractual provisions (Filatotchev et al., 2005). In default situations, VCs also resort to non-legal sanctions and, for instance, exclude non-compliant partners from future syndication networks as a reputation-damaging and economically harmful response in an industry that heavily depends on repeated interaction (Sorenson & Stuart, 2001; Wright & Lockett, 2003). Meuleman et al. (2017) and Zhang et al. (2017) emphasize the importance of prior ties in dealing with the uncertainty surrounding potential partners. Bottazzi et al. (2016) stress the governing effect of trust. We provide an overview of syndication literature in Table 1.

In sum, VC syndication is a trade-off between the resources and benefits that a partner can provide and the risk and uncertainty that the relation with a partner involves (e.g., Brander et al., 2002; Lockett & Wright, 1999; Lockett & Wright, 2001; Manigart et al., 2006; Hopp & Rieder, 2011). Herein, we argue that resources like (international) recognition, which position certain VCs as potential (co-) investors in a closed network structure in this paper, are of particular importance. Blending an international dimension into this trade-off between partner resources and costs further complicates the syndication decision. Specifically, it adds the role of the local institutional context and the distance between the VC's home country context and the target country's institutional setting. In international deals, information asymmetries are amplified by increased geographic, institutional, and cultural distance, which were found to affect foreign investment likelihood (Hain et al., 2016; Moore et al., 2015) and performance (Buchner et al., 2018; Kollmann et al., 2023). Therefore, scholars have examined syndication's effectiveness in addressing distance-related information asymmetry (e.g., Chemmanur et al., 2016; Dai et al., 2012; Tykvova & Schertler, 2014). However, comparatively little is known about the pressing follow-up question of how VC firms form these syndicates in a cross-border context, which is of key relevance, especially for co-investors and entrepreneurs. Two dimensions of syndication are of particular significance because they influence both the costs and benefits: syndicate size and local partners.

Firstly, larger VC syndicates expand the pool of available and non-overlapping resources provided to startups and help secure required investments in subsequent financing rounds. However, adding more co-investors results in exponentially increasing coordination costs, given the higher number of dyadic VC firm relationships (Kim & Park, 2021; Zhang et al., 2017). Prior research indicates dysfunctionalities stemming from larger syndicates are associated with poorer performance (Falconieri et al., 2019). More precisely, the relationship between syndicate size and performance has been found to follow an inverted

U-shape, where larger syndicate size becomes detrimental to investment performance beyond a certain point. This way, medium-sized networks perform better compared to small and large syndicates (Kim & Park, 2021).

Secondly, foreign-local partnering represents a mechanism that helps cross-border investors overcome their proximity disadvantage. Local partners contribute local market knowledge and maintain sufficient personal contact to ensure portfolio companies' effective screening and monitoring (Chemmanur et al., 2016; Hain et al., 2016; Tykvova & Schertler, 2014). Syndicates with local partners reduce information frictions and LOF, increasing the likelihood of investing in informationally opaque firms (e.g., early-stage, high-tech ventures) and successful exits. However, there are also downsides to the diverging nature of foreign and local VC firms. For instance, cultural differences are associated with mutual distrust and ineffective communication, potentially impeding their collaboration (Dai et al., 2012).

2.3. Determinants of syndicate characteristics in cross-border investments

The intricate dynamics of distance between investors and entrepreneurial ventures in cross-border VC flows (e.g., Hain et al., 2016; Moore et al., 2015; Giannetti & Yafeh, 2012) and its impact on investment performance are well recognized in the literature (e.g., Buchner et al., 2018; Li et al., 2014). Syndication, while a promising strategy to address such distance-related issues (e.g., Hain et al., 2016; Tykvova et al., 2014), also introduces a layer of complexity with horizontal agency conflicts among syndicate members. This study delves into the direct effects of two types of within-syndicate distance on syndicate size and local partnering, adding a new layer of complexity to the discussion.

The literature has consistently highlighted the relevance of geographic proximity between investors and investees. We extend this reasoning to the syndicate level, arguing that geographic distance between investing VCs increases agency costs, making the scheduling of regular, joint, in-person meetings with fellow investors more challenging. Moreover, geographic dispersion presumably decreases the likelihood of informal, ad-hoc meetings between partners. Thus, we stress that geographic distance between syndicate members will likely influence syndicate formation, similar to the distance between VCs and the target portfolio company (Dai & Nahata, 2016).

Secondly, we investigate the role of psychic distance between syndicate members. The construct gained considerable attention in international business research following its development by Johanson and Vahlne (1977), who referred to it as 'the sum of factors preventing the flow of information to and from the market.' Macro-level factors such as language, religion, education, political systems, and industrial development inform decision-makers' perceptions of psychic distance (Dow & Karunaratna, 2006). Prior IB research indicates that sizeable psychic distance increases managers' perception of risk due to difficulties in understanding, communicating, and negotiating with stakeholders in the opposing country, which hampers their ability to predict and manage future events (Dow et al., 2020). In the context of VC, Dai & Nahata (2016) find that high cultural distance between home and host markets regularly used to measure psychic distance, is associated with smaller syndicates. We argue that psychic distance is more appropriate to capture partnership decisions in closed networks as they capture decision makers' perceptions of collaborating with a (co-)investor from a specific country, as opposed to cultural and institutional distance measures, which focus more strongly on country differences that impede collaborations with (co-)investors post partnership formation.

In our paper, we assume psychic distance poses a significant source of uncertainty in the context of syndication networks, where it potentially impedes the effectiveness and predictability of the collaboration between syndicate members. This uncertainty might concern disputes regarding individual VCs claiming rights or not redeeming obligations and disagreements on the investee's strategic and operational management. Thus, we reckon the psychic distance between syndicate members

influences syndicate formation.

As Dai & Nahata (2016 p.142) put it 'It thus becomes important from the perspective of foreign investors to decide whether to add one more layer of risk – through syndication with culturally distant investors – to the already multiple risks inherent in cross-border investing.' Our argument is similar but Dai & Nahata (2016) refer to and measure distance between foreign and local VC investors as a reflection of liability of foreignness. Therein, costs of distance is conceptually limited to the presence of a local VC partner.

We argue that distance-related agency-resource trade-offs not only involve local VC partners but also the trade-off to involve other foreign co-investors. This trade-off is conceptually and empirically distinct from the liability of foreignness, as all partners involved may be foreign, but it is nevertheless essential to the ease of collaboration within a (foreign) syndicate. Foreign VCs tend to keep syndicates smaller in case of high geographic and psychic within-syndicate distance to confine agency costs and mitigate uncertainties. In addition, higher within-syndicate distance renders collaborations with local partners more likely to compensate for the syndicates' decelerated and inflexible self-management. This helps ensure effective, on-site monitoring of and unambiguous communication with the portfolio companies and aligns with findings emphasizing local VCs' value in contributing frequent advice on operational issues (Makela & Maula, 2008). We propose the following hypotheses by establishing distance as a significant driver of agency costs.

Hypothesis 1. High geographic distance between syndicate members is associated with (a) smaller syndicates and (b) a higher likelihood of local partners in cross-border VC deals.

Hypothesis 2. High psychic distance between syndicate members is associated with (a) smaller syndicates and (b) a higher likelihood of local partners in cross-border VC deals.

Thus, individual capabilities influence VCs' dependence on and willingness to ally with partners, affecting not only syndication likelihood per se but also how syndicates are formed (Zhang & Pezeshkan, 2016).

The resource-based perspective considers syndication a viable instrument to share and benefit from complimentary capabilities of co-investing partners. Accumulating such resources improves value-adding monitoring activities and, thus also, investment performance (Arundale, 2020; Brander et al., 2002; Lockett & Wright, 2001). RBV, however, focuses commonly on resources and capabilities generated within a company that can be leveraged in economic transactions. Much less attention is devoted to relationship resources like reputation or recognition that provide firms with valuable, rare, and non-imitable advantages in relationship building. We argue that such recognition is particularly important in the closed networks of VC investments and syndication. This is particularly true when VCs internationalize their investment portfolio because VCs must adapt to local market conditions rather than clone their domestic strategies (Wright et al., 2002).

Past studies have studied the role of past investment experience and local host country knowledge (e.g., Meuleman & Wright, 2011; Wang, 2017) as an essential resource shared in VC syndicates. Knowledge is embedded within VCs, allowing the investors to minimize information asymmetry between the VC and the local market. An equally essential yet barely recognized resource is recognition by global and local stakeholders. Recognition reduces the information asymmetry between external stakeholders and the foreign VC. In VC practice, recognition can provide access to more investment opportunities (deal sourcing), help attract local limited partners (fundraising), and open up networking opportunities that can provide value-added to the financed portfolio firms (Bernstein et al., 2016; Hellmann and Puri, 2002). This type of resource has not been studied because most VC literature has falsely portrayed the VC investment decision as a one-sided selection process rather than a two-sided matching process (Dushnitsky and Lenox, 2006;

Hsu, 2004; Wright and Lockett, 2003; Makela & Maula, 2006) in which signaling and recognition play a pivotal role.

We focus on the effects of two types of VCs' recognition on syndicate formation. Firstly, we borrow from the general degree of internationalization and conceptualize a VC firm's general or global recognition, representing the dispersion/diversity of VC activities across markets. Global recognition provides status and can enhance VC firms' access to deal-sourcing channels. Global recognition confers relationship capital and a positive signal to a diverse range of stakeholders (Casillas and Moreno-Menendez, 2014; Zhang et al., 2010) that can be leveraged for the benefit of the VC or the financed portfolio company. Diverse stakeholder access increases comfort in dealing with uncertainty in a foreign market context (De Clercq, Sapienza, & Crijns, 2005). Further, when facing problems in a new domain, such as a new host market, access to diverse stakeholders makes it more likely to find solutions (Gavetti et al., 2005; Meuleman and Wright, 2011). While knowledge lies within the firm, recognition in global markets is equally borne from experience but built up outside of the VC. Like knowledge pollination (Erkelens, Van den Hooff, Huysman & Vlaar, 2015), recognition from dispersed markets may also be transferred into valuable relationship capital in other markets.

Secondly, we follow the concept of country-specific internationalization, or depth of internationalization (Casillas & Moreno-Menendez, 2014), and conceptualize host-country recognition as the VC firm's recognition within a specific VC market. Host market recognition thus creates domestically embedded relationship capital (Erkelens et al., 2015). While global recognition alleviates CMLOF in general, host-country-specific recognition reduces CMLOF in a particular market because the VC does not face information asymmetry from the local market players. Contrary to global, general recognition, local recognition directly applies to the specific local environment and reduces the liabilities of foreignness and the competitive disadvantage in the host market (Delios & Beamish, 2001; Johanson et al., 1977).

We argue that both types of recognition constitute valuable resources in VC internationalization and will affect how VCs internationalize. Recognition provides easier access to ad-hoc resources when needed and reduces the need to co-invest with other VCs, sharing profits. Rather than securing resources through syndication, a recognized VC will invest in smaller syndicates and with fewer local partners relying on the fact that his global recognition will allow access to additional required resources. Therein, recognition by external stakeholders is related to but yet different from experience. While experience creates internal knowledge about how to navigate a specific uncertainty, recognition is a signal that provides privileged access to external resources if needed, irrespective of the content of the VCs own experience. A VC with global or local recognition can count more on accessing required resources when a problem arises and does not rely to equal extent on pooling of resources through syndication. In VC investment both types of recognition, within the target country or the global VC market, can provide VCs with relevant relationships for their operations in a market. This reduces VCs' dependence on co-investors and demand for local relationship capital. Consequently, we expect high degrees of recognition on both levels to lead to smaller syndicates and a reduced likelihood of local partner representation. Thus, we hypothesize as follows.

Hypothesis 3. Intensive host market recognition of the VC is associated with (a) smaller syndicates and (b) a lower likelihood of local partners in cross-border VC deals.

Hypothesis 4. A high degree of general recognition of the VC is associated with (a) smaller syndicates and (b) a lower likelihood of local partners in cross-border VC deals.

3. Data and methodology

3.1. Data sources and sample

The data on VC firm investments is sourced from [Preqin \(2021\)](#). Preqin pioneered the use of FOIA to collect transparent fund return data. With a coverage of 242,466 investments in the VC segment and claimed 99,000 industry practitioners subscribed, it is a dominant data provider in the alternative asset segment, especially with regard to fund performance data. In research, Preqin has been used by leading scholars in PE ([Harris et al., 2014](#); [Lerner et al., 2007](#)). It claims more than 200 citations per year and is a partner of the Wharton Research Data Services (WRDS) ([Preqin, 2024](#)).

In contrast, the scarce studies dealing with syndicate characteristics in international settings (e.g., [Khavul & Deeds, 2016](#); [Liu & Maula, 2016](#); [Meuleman et al., 2017](#)) are based on samples dating back at least to the 2000s, when we saw much lower levels in absolute and relative cross-border activity ([Appendix A](#)). To construct the sample, we start with all syndicated cross-border VC investments in the period from 2010 to 2020, which include syndication-related data to create the dependent variables, location data required to match additional information required for the independent variables, as well as deal-, firm-, and fund-specific information used as controls, yielding 18,964 observations. This time frame marks the period of highest growth in the share of cross-border investments ([Appendix A](#)).

Next, we augment the sample with data required for the independent variables. Firstly, we add within-syndicate distance data regarding geographic distance from [Berry \(2020\)](#) and psychic distance [Dow \(2021\)](#). Secondly, we complement the sample with search data from Google Trends, an easily accessible and vast data source researchers increasingly leverage for diagnosis and forecasting across various disciplines, including business management ([Jun et al., 2018](#)). Thirdly, we add host country data on annual capital market transactions (VC investments, exits of VC-backed firms) retrieved from Preqin and governance data from the World Governance Indicators project by [Kaufmann & Kraay \(2021\)](#).

In our dataset, we have 1,556 unique VC firms, investing in 7,606 portfolio companies in 55 countries. These investments are made by syndicates comprising between 2 and 35 VCs (average of 4 VCs). Because we model individual VC's foreign investments, a syndicated investment of 3 VC funds will create 3 observations ($N = 3$). Our full dataset includes 15,725 VC-level investments. This focal firm perspective reflects our theorizing on the decision of the single VC investing in a foreign country and is in line with comparable studies on alliances and syndicates ([Mindruta et al., 2016](#)). The top five destination countries are the US (24.3 %), China (11.1 %), India (10.5 %), UK (8.7 %), and Canada (5.8 %). The top source countries are the US (40.9 %), the UK (11.5 %), France (6.0 %), Hong Kong (4.4 %), and Germany (3.4 %).

3.2. Methodology

In this study, we conduct regression analyses for the two main dependent variables using generalized linear models (GLM). We run a GLM with a Poisson link function to test the hypotheses on the discrete count data outcome variable syndication size and a GLM with a logit link function to test the hypotheses on the binomial outcome variable likelihood of local partnering.

3.3. Measures

Dependent variables. We construct two measures characterizing syndicate formation in VC deals using data from [Preqin \(2021\)](#). *Syndicate size* is created by summing up the number of VC firms participating in a deal. *The likelihood of local partnering* is created by identifying whether the syndicate includes a VC firm in the same country as the investee (host country), yielding a dichotomous variable. In order to

Table 2
Variable Descriptions.

DV	
Syndicate size	Represents the number of VC firms participating in a deal, based on data retrieved from Preqin.
Likelihood of local partnering	Represents a dichotomous variable, created by identifying whether the syndicate includes a VC firm located in the same country as the investee (host country), based on data retrieved from Preqin.
IV	
Geographic distance	Represents the average <i>geographic</i> distance of all VC firm dyads in a syndicate. We use geographic distance, defined as the great-circle distance between the geographic centers of two countries, leveraging data provided by Berry (2020) . The sum of all dyadic geographic distances in a syndicate is divided by the number of permutations. Only deals with complete distance data are included.
Psychic distance	Represents the average <i>psychic</i> distance of all VC firm dyads in a syndicate. We use the formative index of psychic distance stimuli provided by Dow (2021) , which includes five components: industrial development, education, democracy, language, and religion. The sum of all dyadic psychic distances in a syndicate is divided by the number of permutations. Only deals with complete distance data are included.
Global recognition	Represents a recognition-based measure derived from Google Trends search volume data. It captures how evenly distributed the interest for a keyword is across countries for the focal VC. For this measure the VC firm name is used as a keyword. The search volume for a VC firm is matched to the analyzed investment in terms of time frame (year). Data obtained using globaltrends R package. It delivers the interest in a keyword as search scores, which can be interpreted as the proportion of keyword search volumes on a country or global level compared to control keywords (e.g., Gmail, Maps, Translate, Wikipedia, YouTube), which serve as an approximation of baseline search activity on Google. For the degree of internationalization, the globaltrends R package computes the uniformity of country search scores using the inverted Gini coefficient.
Host market recognition	Represents a recognition-based measure derived from Google Trends search volume data for the focal VC. Depicts the interest for a keyword in a specific market per year. For this measure the VC firm name is used as a keyword. The search volume for a VC firm is matched to the analyzed investment in terms of time frame (year) and host market (country of investee). Data obtained using globaltrends R package. It delivers the interest in a keyword as search scores, which can be interpreted as the proportion of keyword search volumes on a country or global level compared to control keywords (e.g., Gmail, Maps, Translate, Wikipedia, YouTube), which serve as an approximation of baseline search activity on Google.

capture a potential relationship between syndicate size and the presence of a local partner, we include the alternative dependent variable in the respective model as a control. Therein, we capture the endogeneity between the two primary dependent variables, syndicate size and presence of local VC.

Independent variables. Firstly, regarding *within-syndicate distance*, we apply a set of two variables representing the distance of VC firm dyads in a syndicate. We normalize inter-partner distances with the number of partners to avoid size bias, creating an average distance measure. We use geographic distance, defined as the great circle distance between the geographic centers of two countries, leveraging data provided by [Berry \(2020\)](#).

In addition, we use the formative index of psychic distance stimuli provided by [Dow \(2021\)](#), which includes five components: industrial development, education, democracy, language, and religion. Psychic distance is a much broader measure than commonly applied cultural distance measures that focus on specific sociocultural traits (e.g., Individualism) and not essential elements of collaboration such as language, religion, and education (e.g., [Dow & Larimo, 2009](#); [Shenkar, 2001](#)).

Table 3
Descriptives.

	variables	n	mean	SD	min	max
(1)	Syndicate size	15,725	4.524	2.524	2	35
(2)	Local partner	15,725	0.721	0.448	0	1
(3)	Geo distance	15,725	4786.006	3451.599	0	19,836.650
(4)	Psy distance	15,725	3.178	1.895	0	7.911
(5)	VC host recognition	15,725	0.00003	0.0002	0	0.016
(6)	VC global recognition	15,725	0.035	0.065	0	0.407
(7)	VC int. experience	15,725	37.173	71.810	0	526
(8)	VC host experience	15,725	14.353	35.486	0	351
(9)	VC deals host	15,725	2546.016	3036.704	1	8594
(10)	IPO exits host	15,725	32.777	45.187	0	200
(11)	Governance host	15,725	0.048	0.971	−3.002	1.088
(12)	VC size	15,725	789.129	6415.277	0.11	98,583.000
(13)	VC age	15,725	2.216	1.988	0	13
(14)	Fund size	15,725	7.163	10.19	1	96
(15)	Fund age	15,725	16.475	16.33	0	588
(16)	Deal size	15,725	44.798	299.33	0.01	14,000.000

More importantly, psychic distance reflects individuals' perceptions of differences between companies. As such, the measure (and concept) captures (and acknowledges) the biases that individual decision-makers (VC investment officers) have when selecting syndicate partners from different countries (Dow & Larimo, 2009). Thus, psychic distance has more conceptual validity in the partnership choice decision than the commonly used cultural distance concept (and measure), which focuses on population average differences that may impede collaboration once established.

Secondly, we apply related but conceptually different approaches regarding VC firm recognition. We include a foreign VCs' general degree of recognition and host market-specific recognition. The two capture two distinct types of resources in the internationalization process (Casillas & Moreno-Menédez, 2014). The Google Trends-based measure of *general recognition* uses the inverted Gini coefficient of the distribution of search queries for a specific VC fund across all countries globally. *Host market recognition* uses the volume of search queries for a specific VC in the target market specific market. We normalize the country-specific search volume using a combined reference of the most googled search terms: Gmail, Maps, Translate, Wikipedia, and YouTube. Thus, the measure captures how strong the search interest is compared to the reference terms within the country. This makes measures comparable across countries. We obtain the data using the global trends R package guided by the work of Müllner and Puhr (2021).

Using Google search queries to measure the global and host-country reach has advantages over traditional measures. First, Google offers full country coverage. Second and more conceptually, Google allows researchers to measure a VC's reputation in a specific location (or globally). Guler & Guillén (2010) show the importance of VCs' status in the investee company's deal generation and value generation. Reputation can give the VC firm access to more investments and syndicates. Such a reputation or recognition is not solely from past investments in the specific country. Using Google Queries captures the recognition of the VC firm globally and within a single country.

Recognizing the similarities and differences between the two resources, we created both Google-based recognition and experience-based internationalization measures and tested them separately and jointly in robustness tests (Tables 9 and 10). We include Table 2 with detailed descriptions of our main variables of interest.

Control variables. Testing the dependent variable syndicate size, we add our second dependent variable, local partner, as a control and vice versa to control a formal relationship between the two variables of strategic interest. Further, we add a set of control variables regarding VC target country, deal-/investee-, VC fund-, and VC firm characteristics that are consistent with the literature on cross-border VC research in general (e.g., Buchner et al., 2018; Espenlaub, Khurshed, & Mohamed, 2015) and, more specifically, also have implications for syndication.

Syndication is contingent on the institutional environment of the investment (Dorobantu et al., 2020). To capture market attractiveness, we use annual *VC deals* in the host market, and *IPO exits* of VC-backed firms in the host market in line with prior research suggesting investment opportunities and developed stock markets encourage cross-border inflows (Schertler & Tykvova, 2011). In addition, we use a composite index of the world governance indicators (WGI) as a *VC governance quality* variable. The WGI is regularly applied in VC research (e.g., Buchner et al., 2018; Li et al., 2014). The index is created using factor analysis (FACTANAL function in R) to demonstrate individual components' suitability for combination in one factor and to produce Thomson's scores. The analysis yields low uniqueness and high factor loadings above 0.8 across all components, a high explained variance of 85 %, and SS loading of 5.1 for the composite index.

We control for *deal size* (USDm), *fund age* (years), *fund size* (USDm), *firm age* (years), and *firm size* (accumulated number of focal funds). All regressions include fixed effects for an investment round, industry, and investment year. To capture potential spatial patterns of syndication that result from the unique geography of the VC market, we conduct robustness checks using *VC home-country*, *Target country* fixed effects separately and jointly, as well as hierarchical GLM models with VC home-country random intercepts and slopes (Models 15–22).

We include deal size to account for findings indicating VC firms syndicate to access larger deals. The finance motive is especially pronounced among smaller VC firms, so we include firm and fund size characteristics (Manigart et al., 2006). By controlling for the investee industry, we also react to findings suggesting a positive association between industry complexity and syndication (Bayar et al., 2020). Given the higher risk and information asymmetries associated with early-stage investments, we include the deal stage to control for venture-level uncertainty as an alternative syndication driver (Cochrane, 2005; Dai et al., 2012). In addition, we add VC firm and fund age as commonly used proxies for reputation (e.g., Cumming & Dai, 2010; Plagmann & Lutz, 2019), which is not only a key characteristic in the investor-investee relationship (Hsu, 2004) but also facilitates cross-border partnering with other VC firms (Meuleman et al., 2017).

4. Results

4.1. Descriptive statistics

Univariate descriptive statistics are presented in Table 3, and correlations in Table 4. We provide descriptive statistics for our main variables of interest grouped by countries in Appendix C.

Starting with the dependent variables, we find the syndicates in the sample to consist of 4.5 VC firms on average. With a mean of 72 %, a significant share of VC firms relies on local partnering in cross-border

Table 4
Correlations.

variables	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)	(13)	(14)	(15)	(16)
(1) Syndicate size	1															
(2) Local partner	0.213	1														
(3) Geo distance	-0.114	0.127	1													
(4) Psy distance	-0.143	0.198	0.449	1												
(5) VC host recognition	-0.011	0.017	0.047	-0.018	1											
(6) VC global recognition	-0.039	-0.035	0.105	-0.016	0.294	1										
(7) VC int. experience	-0.001	0.023	0.113	-0.006	0.108	0.475	1									
(8) VC host experience	0.007	0.084	0.078	-0.062	0.106	0.292	0.769	1								
(9) VC deals host	0.094	0.247	0.036	-0.079	0.041	-0.1	0.020	0.210	1							
(10) IPO exits host	0.105	0.235	0.04	-0.034	0.038	-0.091	0.053	0.236	0.829	1						
(11) Governance host	0.077	0.09	-0.271	-0.162	-0.032	-0.12	-0.133	-0.150	0.036	-0.107	1					
(12) VC size	0.016	-0.014	0.031	-0.015	-0.004	-0.013	0.011	-0.001	0.034	0.047	-0.035	1				
(13) VC age	0.054	0.02	-0.059	-0.039	0.003	0.0001	-0.007	0.006	0.002	-0.022	0.08	-0.027	1			
(14) Fund size	-0.008	-0.003	0.151	0.046	0.142	0.511	0.679	0.472	-0.007	0.061	-0.234	-0.004	-0.16	1		
(15) Fund age	0.008	-0.015	0.089	-0.003	0.099	0.259	0.349	0.238	-0.051	-0.004	-0.141	-0.018	0.11	0.487	1	
(16) Deal size	0.17	0.016	0.034	0.001	0.024	0.045	0.024	0.026	0.057	0.081	-0.082	0.156	-0.007	0.058	0.036	1

investments. Our key independent variables, geographic and psychic distance, range from 0 to 19,836 km or 7.911 on the psychic distance scale. Within-syndicate distance, the average dyadic geographic distance between syndicate members amounts to 4786 km, corresponding to the geographic distance between the US and Ecuador. The average within-syndicate dyadic psychic distance amounts to 3.2, corresponding to the psychic distance between the US and South Africa. VC recognition ranges from 0 to 0.016 (host country recognition) to 0.407 (general recognition of Sequoia Capital). VC global (local) recognition range from 0 to 0.407 or (0.016). Measures are scaled relative to the most searched terms globally (or in the host country). We argue that recognition is a naturally related but theoretically and empirically distinct resource in VC syndication. Fig. 1 shows the significant correlation of 0.31 ($p = 0.000$) between a VC firm's international investment experience and its global recognition.

The most internationally experienced VC firms are Sequoia, Index Ventures, Accel, 500 Startups and Lightspeed Partners. All of these global funds rank among the Top 10 in their global recognition as measured by Google data. Descriptive statistics of our main variables of interest grouped by VC firm can be found in Appendix 7.

VC market development in host markets, the average number of annual VC deals accounts for 2546, and roughly 33 VC-backed companies exit via public markets per year. Host market governance represents a composite index, which combines six components into a single factor, thereby producing factor scores that are standardized to a mean of zero. The measure serves a holistic proxy for quality of institutional environment. Though the country-level differences are not at the core of our paper, correlations between our main variables of interest and institutional quality provide some interesting insights illustrated in Fig. 2.

While institutional quality of the host country does not seem to influence syndicate sizes, there is a strong and significant correlation between host-country institutional quality and the participation of a local VC syndicate partner. If we conduct a median split of the sample along quality of institutions, mean syndicate size is slightly higher in syndicates investing in target markets with high quality institutions (Appendix D). However, this difference is insignificant.

Regarding control variables, we report an average deal size of \$44.8 m and a fund size of \$789.1 m. Concerning VC firm size, expressed via the cumulative number of funds, on average, investments were conducted by deploying capital from the firm's seventh fund. VC firms in the sample exhibit a mean age of 16.5 years compared to an average fund age of 2.2 years.

Analyzing the sample ($n = 15,725$) from an industry perspective (appendix B), we find 81 % of investments are concentrated in the top three industries: IT (8337), healthcare (2626), and consumer discretionary (1718). Adopting a deal-level perspective, approximately 84 % of investments represent early-stage deals, i.e., angel-, seed-rounds, series A/B, or late-stage deals, i.e., series C to series J, the remaining share pertaining to unspecified rounds or particular types of investments (e.g., secondary stock purchase). We tabulate industry, round, and years in appendices 1&2.

Correlations are presented in Table 4. We follow the best practice recommendation of Lindner et al. (2020) and avoid dropping variables with high correlation (creating omitted variable bias). Instead, we accept inflated standard errors and a more conservative (but unbiased estimation). We also calculate generalized variance inflation factors (GVIF), yielding values below 5, indicating no multicollinearity problems (Hair et al., 1995). A particular concern is the possible correlation between VC experience as a proxy for resources and a VC's need to syndicate. We do not find a significant correlation. International experience is correlated only -0.01 (t-value: -0.58) with syndicate size and international recognition measured with Google is correlated at 0.03 (t-value: 0.99).

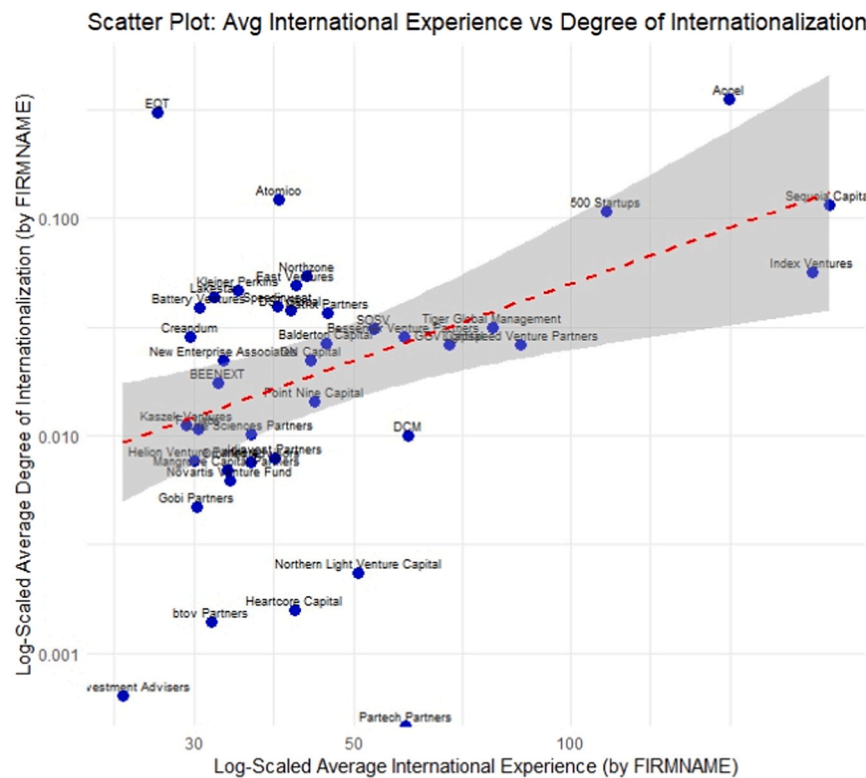


Fig. 1. Scatter Plot International Experience & Recognition.

4.2. Regression results

Tables 5 and 6 report the baseline results for our two dependent variables, syndicate size and local partner. In each regression, we control for the other dependent variable and include year, round, and industry-fixed effects. Model 1 shows the empty model with controls only. Models 2–5 model syndicate size testing H1a to H4a, and Model 6 is the full model with all hypothesized effects jointly. H1a and H2a predict a negative relationship between within-syndicate geographic and psychic distance and syndicate size. The analysis shows negative effects on syndicate size for both types of within-syndicate distance, i.e., geographic distance ($p = 0.0000$) and psychic distance ($p = 0.0000$). Thus, we find support for H1a and H2a.

H3a and H4a suggest a negative relationship between two types of VC host-country recognition and general recognition and syndicate size tested in models 4 and 5. The effects of VC firm host market recognition are negative yet narrowly insignificant ($p = 0.113$). Hence, H3a is not supported by the results. VC firms' degree of recognition is significantly negatively associated with the number of syndicate members ($p = 0.001$), consistent with H4a. The results in the full model 6 support these findings. To test if our recognition findings are related to experience, we test the same regression using only first-time investments where experience is zero in Model 7. Indeed VC global recognition loses its significance indicating that syndicate size may be determined more by experience considerations. Counterfactual models using the restricted sample of repeated investors can be found in appendices E&F. We conduct further analysis on this issue.

Results on control variables are consistent with expectations. The presence of a local partner increases syndicate size significantly. Better governance in target markets facilitates syndication and risk sharing between VCs and creates larger syndicates. Larger VCs can take on more risk and rely less on diversifying risk in syndicates. Consequently, larger VCs have significantly smaller syndicates. On the other hand, deal size increases syndicate size as VCs seek to share the risks of more significant investments.

Local partnering. To test H1b–H4b, we regress the local partnering dependent variable on four hypothesized determinants of syndicate formation in models 9–12 and jointly in model 13 of Table 6. H1b and H2b predict a positive relationship between within-syndicate distance and local partnering, following the rationale that the distant syndicates need a member near the investee to compensate for any distance-related inflexibility of the syndicate. Models 9 and 10 show positive effects on local partnering for the variables on within-syndicate geographic distance ($p = 0.000$) and psychic distance ($p = 0.011$), thus supporting H1b and H2b. H3b and H4b suggest a negative relationship between the two types of VC firm local host market recognition and general recognition and local partnering since we expect that recognition will reduce their need for local partners' market expertise. Model 11 yields contradicting significant effects indicating a significant positive relationship between local host market recognition and local partnering. This surprising finding, we believe, may stem from the two-sided matching. While locally recognized VCs be less dependent on local partners in their investment, as we hypothesize, they may also be more likely to be invited into syndicates by local VCs, due to their recognition (Liu et al., 2016). In model 13, we include only first-time investors. Results on host market recognition are even stronger indicating that host country recognition has a very strong effect on local partnering even if experience is zero.

4.3. Robustness checks

In Tables 7 and 8, we test if our findings may result from geographic patterns in the VC market. We include VC home country fixed effects in models 15 (Table 7 – Syndicate size) and 19 (Table 8 – Local partner) since syndication may differ across VCs from different countries. Results, however, mirror our baseline results, suggesting that our mechanism is generalizable across VCs from different countries. In models 16 (Table 7 – Syndication size) and 20 (Table 8 – Local partner), we include target country fixed effects, recognizing that some target market characteristics may change the fundamentals of the syndication strategy. Again,

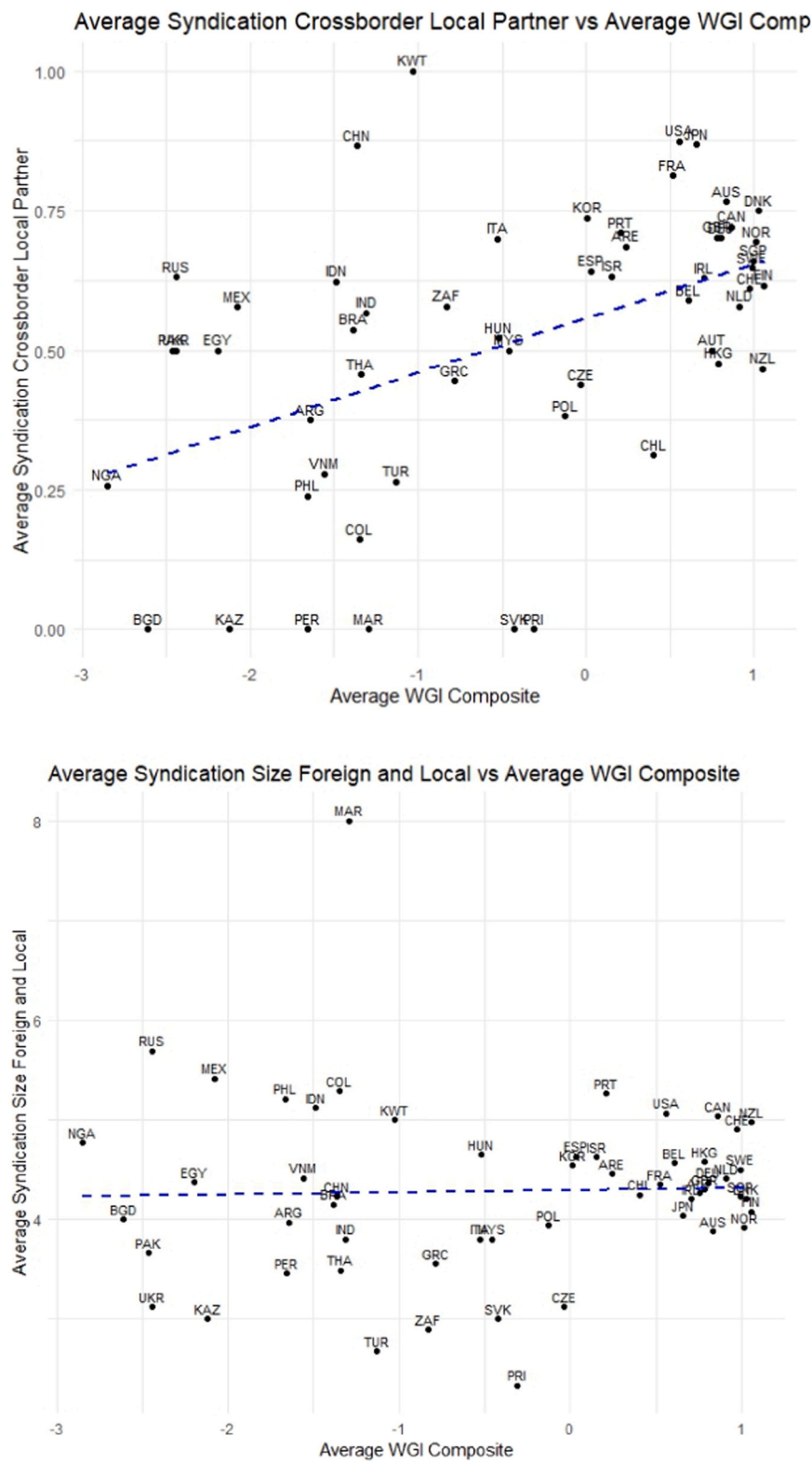


Fig. 2. Scatter Plot Institutional Quality, Local Partners & Syndicate Size.

results mirror our baseline findings closely, suggesting that unobserved target market characteristics do not drive our results. In models 17 (Table 7 – Syndicate size) and 19 (Table 8 – Local Partner), we include both home and host-country FE. Despite a high risk of overfitting, our baseline results broadly hold both significance and magnitude. In models 18 (Table 7 – Syndicate size) and 22 (Table 8 – Local partner), we use a hierarchical binomial GLM with a Poisson link (model 18) and a

binomial link (Model 22) to allow VC-home country random intercepts and random slopes for our hypothesized effects. Again, the results are broadly in line with our baseline findings.

In Tables 9 and 10, we conduct a post-hoc analysis to gain more insights into the role of VC internationalization and the resources underlying its effect on syndication. Specifically, we argue above that our Google-based measure of VC recognition – advantages of data

Table 5
Baseline Models Syndicate Size.

	DV: Syndicate Size						
	1 empty	2 H1a	3 H2a	4 H3a	5 H4a	6 full	7 First time investors
H1a: Geographic distance		−0.00002 *** (0.00000)				−0.00001 *** (0.00000)	−0.00001 *** (0.00000)
H2a: Psychic distance			−0.053 *** (0.002)			−0.045 *** (0.002)	−0.044 *** (0.004)
H3a: VC Host market recognition				−45.989 (28.975)		−24.810 (27.954)	−17.182 (29.180)
H4a: VC global recognition					−0.239 *** (0.071)	−0.269 *** (0.074)	−0.024 (0.167)
Local partner	0.233 *** (0.009)	0.260 *** (0.010)	0.289 *** (0.010)	0.234 *** (0.009)	0.233 *** (0.009)	0.295 *** (0.010)	0.302 *** (0.016)
VC deals host market	0.00000 (0.00000)	0.00001 ** (0.00000)	−0.00000 (0.00000)	0.00000 (0.00000)	0.00000 (0.00000)	0.00000 (0.00000)	0.00000 (0.00000)
VC IPOs host market	0.0002 (0.0001)	−0.00004 (0.0002)	0.0001 (0.0001)	0.0002 (0.0001)	0.0001 (0.0002)	−0.0001 (0.0002)	−0.0004 (0.0003)
VC governance quality	0.053 *** (0.004)	0.032 *** (0.005)	0.032 *** (0.005)	0.053 *** (0.004)	0.053 *** (0.004)	0.025 *** (0.005)	0.001 (0.008)
VC size USDM	−0.00000 *** (0.00000)	−0.00000 *** (0.00000)	−0.00000 *** (0.00000)	−0.00000 *** (0.00000)	−0.00000 *** (0.00000)	−0.00000 *** (0.00000)	−0.00001 *** (0.00000)
VC age	0.003 (0.002)	0.002 (0.002)	0.003 (0.002)	0.003 * (0.002)	0.004 ** (0.002)	0.003 (0.002)	−0.005 (0.004)
Fund size	−0.001 *** (0.0005)	−0.001 ** (0.0005)	−0.001 *** (0.0005)	−0.001 *** (0.0005)	−0.001 (0.001)	−0.0001 (0.001)	−0.0003 (0.001)
Fund age	−0.0003 (0.0003)	−0.0002 (0.0003)	−0.001 * (0.0003)	−0.0003 (0.0003)	−0.0003 (0.0003)	−0.0004 (0.0003)	0.001 (0.0005)
Deal size	0.0001 *** (0.00001)	0.0001 *** (0.00001)	0.0001 *** (0.00001)	0.0001 *** (0.00001)	0.0001 *** (0.00001)	0.0001 *** (0.00001)	0.0005 *** (0.00004)
Constant	1.520 *** (0.355)	1.607 *** (0.355)	1.642 *** (0.355)	1.525 *** (0.355)	1.541 *** (0.355)	1.695 *** (0.355)	0.740 ** (0.368)
Round FE	YES	YES	YES	YES	YES	YES	YES
Industry FE	YES	YES	YES	YES	YES	YES	YES
Year FE	YES	YES	YES	YES	YES	YES	YES
VC Home Country FE	NO	NO	NO	NO	NO	NO	NO
Target Country FE	NO	NO	NO	NO	NO	NO	NO
Observations	15.725	15.725	15.725	15.725	15.725	15.725	4901
Log Likelihood	−32,701.500	−32,539.310	−32,409.460	−32,700.040	−32,695.810	−32,361.810	−10,004.260
Akaike Inf. Crit.	65,483.000	65,160.610	64,900.910	65,482.080	65,473.610	64,811.620	20,094.520

*p < 0.1; **p < 0.05; ***p < 0.01

Table 5: Baseline glm regression models with Poisson family-link.

availability aside – captures an overlooked external resource that is particularly valuable when VCs invest abroad. VC recognition may be accumulated through VC internationalization, but it is conceptually different from internal resources that, like knowledge and experience, VCs may accumulate over the course of their internationalization. In models 23 (Table 9 – Syndicate Size) and 26 (Table 10 – Local Partner), we replace the Google-based measure for general internationalization with the VC firm's general experience in cross-border investment (defined as a count of prior investments not in the VC's home country or the investment target country). In models 24 (Table 9 – Syndicate Size) and 26 (Table 10 – Local Partner), we replace the Google-based host-country score with the VC's specific investment experience in the host country; and in models 25 (Table 9 – Syndicate Size) and 28 (Table 10 – local partner) we replace both.

Results are similar, supporting the notion that both recognition and international experience are valuable resources that (VC) firms accumulate through internationalization. However, there are some nuanced differences in how experience and recognition affect syndication in cross-border investments. Specifically, global recognition decreases syndicate sizes, but empirically, general international experience does not (Model 25). This may indicate that global recognition as a valuable resource is more easily transferable to a specific (but unrelated) VC market context. A similar pattern is also supported in Model 28, in which local partnership is significantly negatively related to host country market experience, indicating that experience in a specific (rather than any) host market reduces the need for local partners. Host country

recognition, on the other hand, is insignificant, indicating that recognition is a less relevant resource locally (than globally).

Endogeneity is an empirical issue in studying syndicates (Dai & Nahata, 2016). Natural experiments are impossible and instruments for the high number of endogenously determined variables like experience, recognition or reputation are imperfect and unavailable. Therefore we follow the recommendation of Wooldridge (2015) for non-linear models to use the control function as a test for endogeneity. Specifically, we regress experience as an independent variable on local partners (syndicate size) in a first stage and include residuals from the first-stage as an independent variable in the second stage with syndicate size (local partner) as a dependent variable. Results (reported in Appendix I) mirror our base-line regressions in terms of coefficients and significancies.

5. Discussion and conclusion

This study contributes to the literature on cross-border VC as a form of international capital mobility, focusing on how firms internationalize concerning determinants of syndicate formation. Therein, it integrates contemporary finance and international business perspectives (Bowe et al., 2010; Puck & Filatotchev, 2018; Puck et al., 2024). VC firms' decision to syndicate seems valid, given prior research suggesting positive performance effects of larger syndicates, at least up to a certain size, and local partners (e.g., Dai et al., 2012; Kim & Park, 2021). However, adding co-investors, local and foreign inflicts additional complexity on cross-border deals. Hence, we provide a nuanced view of

Table 6
Baseline Models Local Partner.

	8 empty	DV: Local Partner					
		9 H1b	10 H2b	11 H3b	12 H4b	13 full	14 First time investors
H1b: Geographic distance		0.0001 *** (0.00001)				0.0001 *** (0.00001)	0.0001 *** (0.00001)
H2b: Psychic distance			0.349 *** (0.011)			0.309 *** (0.012)	0.296 *** (0.020)
H3b: VC Host market recognition				370.474 * (213.236)		356.023 (242.170)	1043.070 ** (462.882)
H4b: VC global recognition					0.254 (0.330)	0.312 (0.385)	−1.151 (0.875)
Syndicate size	0.280 *** (0.012)	0.304 *** (0.012)	0.332 *** (0.012)	0.281 *** (0.012)	0.280 *** (0.012)	0.341 *** (0.012)	0.353 *** (0.021)
VC deals host market	0.0001 *** (0.00001)	0.0001 *** (0.00001)	0.0002 *** (0.00002)	0.0001 *** (0.00001)	0.0001 *** (0.00001)	0.0001 *** (0.00002)	0.0001 *** (0.00003)
VC IPOs host market	0.008 *** (0.001)	0.009 *** (0.001)	0.007 *** (0.001)	0.008 *** (0.001)	0.008 *** (0.001)	0.008 *** (0.001)	0.005 *** (0.002)
VC governance quality	0.184 *** (0.020)	0.304 *** (0.021)	0.287 *** (0.022)	0.185 *** (0.020)	0.184 *** (0.020)	0.333 *** (0.022)	0.388 *** (0.038)
VC size USDM	−0.00000 (0.00000)	−0.00001 * (0.00000)	−0.00000 (0.00000)	−0.00000 (0.00000)	−0.00000 (0.00000)	−0.00000 (0.00000)	−0.00000 (0.00001)
VC age	0.010 (0.010)	0.015 (0.010)	0.008 (0.011)	0.009 (0.010)	0.009 (0.010)	0.009 (0.011)	−0.013 (0.019)
Fund size	0.004 ** (0.002)	0.001 (0.002)	0.004 * (0.002)	0.004 (0.002)	0.004 (0.003)	0.0004 (0.003)	−0.009 (0.007)
Fund age	−0.001 (0.001)	−0.002 (0.001)	0.001 (0.001)	−0.001 (0.001)	−0.001 (0.001)	0.0004 (0.001)	−0.002 (0.002)
Deal size	−0.0002 ** (0.0001)	−0.0002 *** (0.0001)	−0.0003 *** (0.0001)	−0.0002 *** (0.0001)	−0.0002 *** (0.0001)	−0.0003 *** (0.0001)	−0.001 * (0.0003)
Constant	12.805 (882.743)	12.014 (882.743)	11.363 (882.743)	12.763 (882.743)	12.780 (882.743)	11.066 (882.743)	13.091 (1455.398)
Round FE	YES	YES	YES	YES	YES	YES	
Industry FE	YES	YES	YES	YES	YES	YES	
Year FE	YES	YES	YES	YES	YES	YES	
VC Home Country FE	NO	NO	NO	NO	NO	NO	
Target Country FE	NO	NO	NO	NO	NO	NO	
Observations	15.725	15.725	15.725	15.725	15.725	15.725	
Log Likelihood	−8198.973	−7958.197	−7652.032	−8196.769	−8198.676	−7603.798	
Akaike Inf. Crit.	16,477.950	15,998.390	15,386.060	16,475.540	16,479.350	15,295.590	

*p < 0.1; **p < 0.05; ***p < 0.01

Table 6: Baseline glm regression models with binomial family-link.

how VC firms invest across borders about syndication by investigating which circumstances influence VC syndicate composition.

VC firms consider deal-specific drivers of agency costs, such as distance. Specifically, we argue that VCs' syndication decision is influenced strongly by psychic distance as psychic distance captures the anticipated decision-makers' perceptions about collaborating with a VC from another country. Further, we argue that VCs rely strongly on recognition and social signaling to overcome such psychic distance in closed networks like VC investments. In addition to anticipated agency costs from distance, VCs contrast value gains from peer contributions against their own resource position. We argue that both general and host-country specific recognition of a focal VC have a considerable influence on this trade-off that is distinct from experience. Experience creates internal knowledge that is highly contextual and resides within a VC. If a VC incurs a challenge, it is uncertain if knowledge from past experience addresses the exact needs to solve the situation. Recognition, or reputation, on the other hand, constitutes a signal towards external stakeholders that can be deployed ad-hoc. Globally and locally recognized VC have distinct but privileged access to external resources beyond their own knowledge and experience.

Knowing how VCs syndicate is of interest to both potential partner VCs and entrepreneurs. Given that foreign and domestic partners contribute in different ways to the success of investees (Makela & Maula, 2006; Makela & Maula, 2008) and larger syndicates might have a higher capacity to put together larger investment rounds in line with the finance motive for syndication (e.g., Gottschalg & Gerasymenko, 2008; Liu & Maula, 2016), entrepreneurs will be interested in the underlying

mechanisms rendering specific, desired syndicate configurations more viable. From the perspective of the internationalizing VC firm or prospective co-investors, it is likewise of interest to follow a set of easily observable and guiding criteria when deciding on syndicate formation. After all, typical advantages of syndication (e.g., sharing of capital infusions, monitoring costs, and required capabilities) might reduce the urgent need for timely exits (Sethuram et al., 2021) or positively influence performance (Chemmanur et al., 2016; Dai et al., 2012). However, the focal syndicate's ability to capitalize on these advantages hinges on its specific setup (Kim & Park, 2021). Hence, this study takes one step back and investigates the determinants of syndicate configurations concerning determinants of key syndicate characteristics: syndicate size and local partnering.

First, the findings on deal-specific distance between the members of a particular syndicate shed light on agency issues and attach importance to the value-added motive concerning cross-organizational collaboration. The importance of proximity for VC firms to effectively monitor and further develop investees is well recognized (e.g., Chemmanur et al., 2016). This study focuses on the distance between syndicate members and shows that higher within-syndicate distance is associated with smaller syndicates. VC firms react to increased complexities incurred via higher geographic and psychic distance between syndicate members by curtailing the number of co-investors. In addition, higher-within syndicate distance is associated with a higher likelihood of partnering with local VC firms.

Second, the effects of recognition depend on their specific nature. General (global) recognition, or reputation, reduces the need to include

Table 7
Robustness checks I.

	DV: Syndicate Size			
	15 VC Home Country FE	16 Target Country FE	17 VC Home & Target Country FE	18 Hierarchical GLM (Poisson)
H1a: Geographic distance	−0.00001 *** (0.00000)	−0.00001 *** (0.00000)	−0.00001 *** (0.00000)	−0.00001 *** (0.00000)
H2a: Psychic distance	−0.049 *** (0.003)	−0.044 *** (0.003)	−0.048 *** (0.003)	−0.046 *** (0.003)
H3a: VC Host market recognition	−18.822 (27.063)	−22.072 (27.160)	−17.651 (26.023)	−32.484 (30.857)
H4a: VC global recognition	−0.130 * (0.077)	−0.264 *** (0.075)	−0.170 ** (0.078)	−0.200 * (0.117)
Local partner	0.307 *** (0.010)	0.309 *** (0.010)	0.321 *** (0.010)	0.300 *** (0.010)
VC deals host market	0.00000 (0.00000)	−0.00000 (0.00000)	−0.00000 (0.00000)	0.00000 (0.00000)
VC IPOs host market	0.0001 (0.0002)	0.001 *** (0.0002)	0.0005 ** (0.0002)	−0.00000 (0.0002)
VC governance quality	0.022 *** (0.005)	0.150 *** (0.040)	0.144 *** (0.040)	0.013 ** (0.005)
VC size USDm	−0.00000 *** (0.00000)	−0.00000 *** (0.00000)	−0.00000 *** (0.00000)	−0.00000 * (0.00000)
VC age	0.005 ** (0.002)	0.003 (0.002)	0.005 *** (0.002)	0.005 ** (0.002)
Fund size	−0.001 ** (0.001)	0.001 (0.001)	−0.0005 (0.001)	−0.0001 (0.001)
Fund age	−0.001 ** (0.0003)	−0.00001 (0.0003)	−0.0003 (0.0003)	−0.0002 (0.0004)
Deal size	0.0001 *** (0.00001)	0.0001 *** (0.00001)	0.0001 *** (0.00001)	0.0001 *** (0.00001)
Constant	1.738 *** (0.360)	2.112 *** (0.373)	2.119 *** (0.378)	1.549 *** (0.377)
Round FE	YES	YES	YES	YES
Industry FE	YES	YES	YES	YES
Year FE	YES	YES	YES	YES
VC Home Country FE	YES	NO	YES	NO
Target Country FE	NO	YES	YES	NO
Observations	15.725	15.725	15.725	15.725
Log Likelihood	−32,203.830	−32,116.770	−31,972.340	−32,207.800
Akaike Inf. Crit.	64,627.670	64,429.530	64,272.690	64,505.600
Bayesian Inf. Crit.				64,850.440

*p < 0.1; **p < 0.05; ***p < 0.01

Table 7: Robustness checks with VC home country fixed effects (15), Target country fixed effects (16), both VC and Target country fixed effects (17) and a hierarchical glm model (glmer) with fund-level random intercepts and slopes.

additional resources in your investment syndicate a-priori as globally recognized VCs can count on privileged access to external resources if needed. This reduces the need for larger syndicates. The effect of host market recognition differs. Our findings suggest that host market recognition improves access to local partners within the partner selection process. This reduces the costs of finding local syndicate partners and in consequence increases the probability of a local partner in the syndicate.

This study makes several contributions that are relevant to academia and practitioners. Regarding implications for the scientific community, we extend the growing body of knowledge on cross-border VC investing by describing how VC firms execute deals and identifying a three-dimensional framework of determinants that influence both syndication size and local partnering. VC firms base their co-investment structure on distance-related agency costs and capability-based resources. Most importantly, our study highlights the importance of signalling through recognition in otherwise closed VC networks, when VC funds seek investments across borders.

Furthermore, this study adds to research on international management, demonstrating the legitimacy of recognition-based measures to capture firms' degree of internationalization as a resource proxy. This is especially valuable in a context where accounting-based measures are unavailable, such as, for instance, in private market analyses. With regard to practitioner insights, our findings are relevant, especially for

three groups of stakeholders. Insights into their investment partners' collaboration motives provide VC firms with a sound basis to make informed syndication decisions. This involves, for instance, having realistic expectations about the necessity of syndicating when entering attractive VC markets and reaching out to adequate partners regarding distance and capabilities when structuring the syndicate. In addition, entrepreneurs are in a better position to plan their funding rounds, knowing that the participation of a local VC is instrumental in syndicates of mutually distant VC firms. However, it might be detrimental to winning over truly internationalized VC firms. Furthermore, findings on the importance of VC market development for syndication (i.e., investments, exits, governance) provide policymakers with levers to lure additional VC flows into the country.

6. Limitations and future research

While we managed to obtain extensive data on cross-border VC transactions, accessing firm-level data on private market investors and portfolio companies remains a considerable challenge and holds opportunities for future research on syndication. In any empirical setting, network formation is characterized by path dependencies, various steps, and a high endogeneity. While we try to reduce endogeneity concerns, including information on past investments, a restricted sample regression, control function regressions (Wooldridge, 2015) creating an

Table 8
Robustness Checks II.

	DV: Local partner			
	19 VC Home Country FE	20 Target Country FE	21 VC Home & Target Country FE	22 Hierarchical GLM (binomial)
H1b: Geographic distance	0.00004 *** (0.00001)	0.0001 *** (0.00001)	0.0001 *** (0.00001)	0.0001 *** (0.00001)
H2b: Psychic distance	0.392 *** (0.015)	0.349 *** (0.014)	0.413 *** (0.016)	0.368 *** (0.015)
H3b: VC Host market recognition	402.228 (253.968)	354.152 (234.393)	270.079 (200.932)	405.994 *** (20.627)
H4b: VC global recognition	-0.207 (0.399)	0.652 * (0.396)	0.309 (0.400)	0.002 (0.652)
Syndicate size	0.353 *** (0.013)	0.370 *** (0.013)	0.383 *** (0.013)	0.379 *** (0.014)
VC deals host market	0.0002 *** (0.00002)	0.0001 *** (0.00003)	0.0001 *** (0.00003)	0.0001 *** (0.00002)
VC IPOs host market	0.007 *** (0.001)	0.002 (0.001)	0.002 * (0.001)	0.009 *** (0.001)
VC governance quality	0.396 *** (0.025)	0.630 *** (0.216)	0.686 *** (0.220)	0.395 *** (0.028)
VC size USDm	-0.00001 (0.00000)	-0.00000 (0.00000)	-0.00001 (0.00000)	-0.00001 (0.00001)
VC age	0.014 (0.011)	0.011 (0.011)	0.010 (0.012)	0.003 (0.014)
Fund size	0.003 (0.003)	-0.003 (0.003)	0.002 (0.003)	-0.002 (0.005)
Fund age	-0.0002 (0.001)	-0.001 (0.001)	-0.0005 (0.001)	0.001 (0.002)
Deal size	-0.0003 *** (0.0001)	-0.0003 *** (0.0001)	-0.0003 *** (0.0001)	-0.0003 *** (0.0001)
Constant	11.167 (1455.397)	11.076 (1455.398)	10.104 (1455.398)	11.556 (15.950)
Round FE	YES	YES	YES	YES
Industry FE	YES	YES	YES	YES
Year FE	YES	YES	YES	YES
VC Home Country FE	YES	NO	YES	NO
Target Country FE	NO	YES	YES	NO
Observations	15.725	15.725	15.725	15.725
Log Likelihood	-7392.936	-7313.316	-7128.510	-7416.426
Akaike Inf. Crit.	15,005.870	14,822.630	14,585.020	14,922.850
Bayesian Inf. Crit.				15,267.690

*p < 0.1; **p < 0.05; ***p < 0.01

Table 8: Robustness checks with VC home country fixed effects (19), Target country fixed effects (20), both VC and Target country fixed effects (21) and a hierarchical glm model (glmer) with fund-level random intercepts and slopes.

empirical model capable of capturing all aspects of network evolution or truly exogenous instruments is unrealistic (Dai & Nahata, 2016). Future studies, therefore, should dig deeper into possible aspects of the syndication process that may create endogeneity (e.g., the role of the lead investors, staging of co-investment, and selection effects on the target company side).

Second, despite our use of one of the most comprehensive datasets on VC investment (Harris et al., 2014; Lerner et al., 2007), some potentially omitted variables could confound our results but also provide for future

Table 9
Robustness checks III.

	DV: Syndicate Size		
	23 Alt.M easure Int.	24 Alt. Measure Host Int.	25 Both Alt. Measures
H1a: Geographic distance	-0.00001 *** (0.00000)	-0.00001 *** (0.00000)	-0.00001 *** (0.00000)
H2a: Psychic distance	-0.045 *** (0.002)	-0.046 *** (0.002)	-0.046 *** (0.002)
H4a: VC international experience	-0.0003 *** (0.0001)		-0.00001 (0.0001)
H3a: VC Host market recognition	-53.872 * (29.876)		
H4a VC global recognition		-0.256 *** (0.071)	
H3a: VC Host market experience		-0.001 *** (0.0001)	-0.001 *** (0.0002)
Local partner	0.296 *** (0.010)	0.298 *** (0.010)	0.298 *** (0.010)
VC deals host market	0.00000 (0.00000)	0.00000 (0.00000)	0.00000 (0.00000)
VC IPOs host market	-0.00003 (0.0002)	-0.00001 (0.0002)	0.00003 (0.0002)
VC governance quality	0.025 *** (0.005)	0.023 *** (0.005)	0.023 *** (0.005)
VC size USDm	-0.00000 *** (0.00000)	-0.00000 *** (0.00000)	-0.00000 *** (0.00000)
VC age	0.004 * (0.002)	0.004 ** (0.002)	0.004 * (0.002)
Fund size	0.0004 (0.001)	0.001 (0.001)	0.0001 (0.001)
Fund age	-0.0003 (0.0003)	-0.0004 (0.0003)	-0.0004 (0.0003)
Deal size	0.0001 *** (0.00001)	0.0001 *** (0.00001)	0.0001 *** (0.00001)
Constant	1.659 *** (0.355)	1.676 *** (0.355)	1.651 *** (0.355)
Round FE	YES	YES	YES
Industry FE	YES	YES	YES
Year FE	YES	YES	YES
VC Home Country FE	NO	NO	NO
Target Country FE	NO	NO	NO
Observations	15.725	15.725	15.725
Log Likelihood	-32,359.990	-32,348.200	-32,354.700
Akaike Inf. Crit.	64,807.980	64,784.410	64,797.390

*p < 0.1; **p < 0.05; ***p < 0.01

Table 9: Robustness checks with alternative measures for VC internationalization variables. In model 23, we replace the Google Trends score for VC fund internationalization with a count variable of a VC fund's investments outside of both the home market and the target market prior to the deal. In model 24, we replace the VC funds host country score from Google Trends with prior investments of the VC in the specific target host country. In model 25, we replace both primary measures with their alternatives.

research opportunities. For example, we cannot observe local VC offices (or offices of target companies in VC home countries) in the data. Representative offices could increase deal flow in a particular country and create the recognition we study in this paper. To assess the likelihood of omitted variable bias, we collected data on the largest VC investors in our sample (Sequoia, Index Ventures, Accel). The three largest funds make up 1.375 investments in our sample (almost 10 %). In between them, they have four foreign offices (US, Israel, UK, India and China) but invested in more than 31 countries (Appendix 8). For the top 5 funds, that make up 12 percent of the investments (1827), the number of foreign offices increases to only 6 (Israel, UK, India, Germany, Singapore, China). These numbers indicate that local offices are the

Table 10
Robustness checks IV.

	DV: Local Partner		
	26 Alt. Measure Int.	27 Alt. Measure Host Int.	28 Both Alt. Measures
H1b: Geographic distance	0.00002 *** (0.00000)	0.00002 *** (0.00000)	0.00002 *** (0.00000)
H2b: Psychic distance	0.082 *** (0.006)	0.084 *** (0.006)	0.083 *** (0.006)
H4b: VC international experience	0.001 *** (0.0002)		−0.0002 (0.0003)
H3b: VC Host market recognition	42.875 (47.342)		
H4b VC global recognition		0.062 (0.180)	
H3b: VC host country experience		0.001 *** (0.0003)	0.002 *** (0.0004)
Local partner	0.055 *** (0.004)	0.055 *** (0.004)	0.055 *** (0.004)
VC deals host market	0.00002 *** (0.00001)	0.00002 *** (0.00001)	0.00002 *** (0.00001)
VC IPOs host market	0.002 *** (0.0004)	0.002 *** (0.0004)	0.001 *** (0.0004)
VC governance quality	0.098 *** (0.012)	0.101 *** (0.012)	0.102 *** (0.012)
VC size US\$M	−0.00000 (0.00000)	−0.00000 (0.00000)	−0.00000 (0.00000)
VC age	0.002 (0.005)	0.002 (0.005)	0.003 (0.005)
Fund size	−0.002 (0.001)	−0.002 (0.001)	−0.001 (0.001)
Fund age	−0.0002 (0.001)	−0.0002 (0.001)	−0.0002 (0.001)
Deal size	−0.00004 (0.00003)	−0.00004 (0.00003)	−0.00004 (0.00003)
Constant	−0.650 (1.006)	−0.653 (1.006)	−0.652 (1.006)
Round FE	YES	YES	YES
Industry FE	YES	YES	YES
Year FE	YES	YES	YES
VC Home Country FE	NO	NO	NO
Target Country FE	NO	NO	NO
Observations	15,725	15,725	15,725
Log Likelihood	−14,631.840	−14,625.850	−14,625.590
Akaike Inf. Crit.	29,351.680	29,339.700	29,339.180

*p < 0.1; **p < 0.05; ***p < 0.01

Table 10 Robustness checks with alternative measures for VC internationalization variables. In model 26, we replace the Google Trends score for VC fund internationalization with a count variable of a VC fund's investments outside of both the home market and the target market prior to the deal. In model 27, we replace the VC funds host country score from Google Trends with prior investments of the VC in the specific target host country. In model 28, we replace both primary measures with their alternatives.

exception and, likely, weaker proxies for location specific resources than prior investment experience, or, as we propose, local recognition. Nevertheless, future research should study the benefits of local offices in VC syndication.

Third, continued research could contrast our findings on the influence of country-level governance on syndicate formation with firm-level governance (e.g., board representation of syndicate members), which might serve as a substitute for relationship capital institutions ensuring contract enforcement.

Fourth, we only analyze the recognition of the investing firm, not the recognition of the investee or the comparative recognition across all partners. Theorizing on and empirically testing a theory of comparative

reputation in a syndicate could be an interesting extension to the literature.

Fifth, scholars could examine alternative options to local partnering as a mechanism to compensate for inefficient syndicate setups. This includes, for instance, a substantial shift of monitoring activities to on-line communication channels in the post-COVID-19 era.

Finally, future research could investigate the effects of VC firm internationalization on syndicate formation on an individual level, focusing on the educational-, occupational-, and investment experience of VC firms' employees managing the focal investments.

Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at [doi:10.1016/j.ibusrev.2025.102424](https://doi.org/10.1016/j.ibusrev.2025.102424).

Data Availability

The authors do not have permission to share data.

References

- Alhorh, H. S., Moore, C. B., & Payne, G. T. (2008). The impact of economic integration on cross-border venture capital investments: Evidence from the European Union. *Entrepreneurship Theory and Practice*, 32(5), 897–917.
- Alvarez-Garrido, E., & Guler, I. (2018). Status in a strange land? Context-dependent value of status in cross-border venture capital. *Strategic Management Journal*, 39(7), 1887–1911.
- Amit, R., Brander, J., & Zott, C. (1998). Why do venture capital firms exist? Theory and Canadian evidence. *Journal of Business Venturing*, 13(6), 441–466.
- Arundale, K. (2020). Syndication and cross-border collaboration by venture capital firms in Europe and the USA: A comparative study. *Venture Capital*, 22(4), 355–376.
- Bayar, O., Chemmanur, T. J., & Tian, X. (2020). Peer monitoring, syndication, and the dynamics of venture capital interactions: theory and evidence. *Journal of Financial and Quantitative Analysis*, 55(6), 1875–1914.
- Bell, R. G., Filatotchev, I., & Rasheed, A. A. (2012). The liability of foreignness in capital markets: Sources and remedies. *Journal of International Business Studies*, 43(2), 107–122.
- Bernstein, S., Giroud, X., & Townsend, R. R. (2016). The impact of venture capital monitoring. *The Journal of Finance*, 71(4), 1591–1622.
- Berry, H. 2020. Cross National Distance Data. (<https://professorheatherberry.weebly.com/>). August 16, 2021.
- Bertoni, F., & Groh, A. P. (2014). Cross-border investments and venture capital exits in Europe. *Corporate Governance-an International Review*, 22(2), 84–99.
- Bottazzi, L., Da Rin, M., & Hellmann, T. (2016). The importance of trust for investment: Evidence from venture Capital. *Review of Financial Studies*, 29(9), 2283–2318.
- Bowe, M., Filatotchev, I., & Marshall, A. (2010). Integrating contemporary finance and international business research. *International Business Review*, 19(5), 435–445.
- Brander, J. A., Amit, R., & Antweiler, W. (2002). Venture-capital syndication: Improved venture selection vs. the value-added hypothesis. *Journal of Economics Management Strategy*, 11(3), 423–452.
- Buchner, A., Espenlaub, S., Khurshed, A., & Mohamed, A. (2018). Cross-border venture capital investments: The impact of foreignness on returns. *Journal of International Business Studies*, 49(5), 575–604.
- Casillas, J. C., & Moreno-Mendez, A. M. (2014). Speed of the internationalization process: the role of diversity and depth in experiential learning. *Journal of International Business Studies*, 45, 85–101.
- Chahine, S., Saade, S., & Goergen, M. (2019). Foreign business activities, foreignness of the VC syndicate, and IPO value. *Entrepreneurship Theory and Practice*, 43(5), 947–973.
- Chemmanur, T. J., Hull, T. J., & Krishnan, K. (2016). Do local and international venture capitalists play well together? The complementarity of local and international venture capitalists. *Journal of Business Venturing*, 31(5), 573–594.
- Cochrane, J. H. (2005). The risk and return of venture capital. *Journal of Financial Economics*, 75(1), 3–52.
- Cumming, D. (2006). Adverse selection and capital structure: evidence from venture capital. *Entrepreneurship Theory and Practice*, 30(2), 155–183.
- Cumming, D., & Dai, N. (2010). Local bias in venture capital investments. *Journal of Empirical Finance*, 17(3), 362–380.
- Cumming, D., Knill, A., & Syvrud, K. (2016). Do international investors enhance private firm value? Evidence from venture capital. *Journal of International Business Studies*, 47, 347–373.
- Dai, N., Jo, H., & Kassicheh, S. (2012). Cross-border venture capital investments in Asia: Selection and exit performance. *Journal of Business Venturing*, 27(6), 666–684.
- Dai, N., & Nahata, R. (2016). Cultural differences and cross-border venture capital syndication. *Journal of International Business Studies*, 47(2), 140–169.
- Dalziel, T., Gentry, R. J., & Bowerman, M. (2011). An integrated agency-resource dependence view of the influence of directors' human and relational capital on firms' R&D spending. *Journal of Management Studies*, 48(6), 1217–1242.

- De Clercq, D., Sapienza, H. J., & Crijns, H. (2005). The internationalization of small and medium-sized firms. *Small Business Economics*, 24(4), 409–419.
- De Priecker, S., Manigart, S., Wright, M., & De Maeseneire, W. (2012). The influence of experiential, inherited and external knowledge on the internationalization of venture capital firms. *International Business Review*, 21(5), 929–940.
- Delios, A., & Beamish, P. W. (2001). Survival and profitability: The roles of experience and intangible assets in foreign subsidiary performance. *Academy of Management Journal*, 44(5), 1028–1038.
- Devigne, D., Manigart, S., Vanacker, T., & Mulier, K. (2018). Venture capital internationalization: Synthesis and future research directions. *Journal of Economic Surveys*, 32(5), 1414–1445.
- Dorobantu, S., Lindner, T., & Müllner, J. (2020). Political risk and alliance diversity: A two-stage model of partner selection in multipartner alliances. *Academy of Management Journal*, 63(6), 1775–1806.
- Dow, D. 2021. Distance and Diversity Scales for International Business Research. (<http://dow.net.au/>). Oktober 1, 2021.
- Dow, D., Baack, D., & Parente, R. (2020). The role of psychic distance in entry mode decisions: Magnifying the threat of opportunism or increasing the need for local knowledge? *Global Strategy Journal*, 10(2), 309–334.
- Dow, D., & Karunaratna, A. (2006). Developing a multidimensional instrument to measure psychic distance stimuli. *Journal of International Business Studies*, 37(5), 578–602.
- Dow, D., & Larimo, J. (2009). Challenging the conceptualization and measurement of distance and international experience in entry mode choice research. *Journal of International Marketing*, 17(2), 74–98.
- Dushnitsky, G., & Lenox, M. J. (2006). When does corporate venture capital investment create firm value? *Journal of Business Venturing*, 21(6), 753–772.
- Espenlaub, S., Khurshed, A., & Mohamed, A. (2015). Venture capital exits in domestic and cross-border investments. *Journal of Banking Finance*, 53, 215–232.
- Erkelens, R., Van den Hooff, B., Huysman, M., & Vlaar, P. (2015). Learning from locally embedded knowledge: Facilitating organizational learning in geographically dispersed settings. *Global Strategy Journal*, 5, 177–197.
- Falconieri, S., Filatotchev, I., & Tastan, M. (2019). Size and diversity in VC syndicates and their impact on IPO performance. *The European Journal of Finance*, 25(11), 1032–1053.
- Filatotchev, I., Chahine, S., Wright, M., & Arberk, M. (2005). Founders' characteristics, venture capital syndication and governance in entrepreneurial IPOs. *The International Entrepreneurship and Management Journal*, 1(4), 419–439.
- Filatotchev, I., Wright, M., & Arberk, M. (2006). Venture capitalists, syndication and governance in initial public offerings. *Small Business Economics*, 26(4), 337–350.
- Fu, H., Yang, J., & An, Y. B. (2019). Made for each other: Perfect matching in venture capital markets. *Journal of Banking Finance*, 100, 346–358.
- Gavetti, G., Levinthal, D. A., & Rivkin, J. W. (2005). Strategy making in novel and complex worlds: The power of analogy. *Strategic Management Journal*, 26(8), 691–712.
- Giannetti, M., & Yafeh, Y. (2012). Do cultural differences between contracting parties matter? Evidence from syndicated bank loans. *Management Science*, 58(2), 365–383.
- Gottschalg, O., & Gerasymenko, V. (2008). *Antecedents and Consequences of Venture Capital Syndication*. (<https://ssrn.com/abstract=1095033>).
- Gu, Y., Filatotchev, I., Bell, R. G., & Rasheed, A. A. (2019). Liability of foreignness in capital markets: Institutional distance and the cost of debt. *Journal of Corporate Finance*, 57, 142–160.
- Guler, I., & Guillén, M. F. (2010). Institutions and the internationalization of US venture capital firms. *Journal of International Business Studies*, 41(2), 185–205.
- Guler, I., & Guillén, M. (2010). Home country networks and foreign expansion: Evidence from the venture capital industry. *Academy of Management Journal*, 53(2), 390.
- Hain, D., Johan, S., & Wang, D. J. (2016). Determinants of cross-border venture capital investments in emerging and developed economies: The effects of relational and institutional trust. *Journal of Business Ethics*, 138(4), 743–764.
- Hair, J. F., Anderson, R. E., Tatham, R. L., & Black, W. C. (1995). *Multivariate data analysis* (4th ed.). with readings: Prentice-Hall, Inc.
- Harris, R. S., Jenkinson, T., & Kaplan, S. N. (2014). Private equity performance: What do we know? *The Journal of Finance*, 69(5), 1851–1882.
- Hellmann, T., & Puri, M. (2002). Venture capital and the professionalization of start-up firms: Empirical evidence. *Journal of Finance*, 57(1), 169–197.
- Hochberg, Y. V., Lindsey, L. A., & Westerfield, M. M. (2015). Resource accumulation through economic ties: Evidence from venture capital. *Journal of Financial Economics*, 118(2), 245–267.
- Hochberg, Y. V., Ljungqvist, A., & Lu, Y. (2007). Whom you know matters: Venture capital networks and investment performance. *The Journal of Finance*, 62(1), 251–301.
- Hochberg, Y. V., Ljungqvist, A., & Lu, Y. (2010). Networking as a barrier to entry and the competitive supply of venture capital. *Journal of Finance*, 65(3), 829–859.
- Hopp, C., & Rieder, F. (2011). What drives venture capital syndication? *Applied Economics*, 43(23), 3089–3102.
- Hsu, D. H. (2004). What do entrepreneurs pay for venture capital affiliation? *Journal of Finance*, 59(4), 1805–1844.
- Huang, X., Kenney, M., & Patton, D. (2015). Responding to uncertainty: syndication partner choice by foreign venture capital firms in China. *Venture Capital*, 17(3), 215–235.
- Jääskeläinen, M. (2012). Venture capital syndication: Synthesis and future directions. *International Journal of Management Reviews*, 14(4), 444–463.
- Johanson, J., & Vahlne, J.-E. (1977). The internationalization process of the firm—A model of knowledge development and increasing foreign market commitments. *Journal of International Business Studies*, 8(1), 23–32.
- Jun, S.-P., Yoo, H. S., & Choi, S. (2018). Ten years of research change using Google Trends: From the perspective of big data utilizations and applications. *Technological Forecasting and Social Change*, 130, 69–87.
- Kaufmann, D., & Kraay, A. 2021. Worldwide Governance Indicators (WGI). <https://info.worldbank.org/governance/wgi/>. September 2021, World Bank.
- Khavul, S., & Deeds, D. (2016). The evolution of initial co-investment syndications in an emerging venture capital market. *Journal of International Management*, 22(3), 280–293.
- Kim, J. Y., & Park, H. D. (2021). The influence of venture capital syndicate size on venture performance. *Venture Capital*, 23(2), 179–203.
- Kollmann, H., Müllner, J., & Puck, J. (2023). Pioneering excellence or fleeing mediocrity? Why venture capital firms internationalize. *Global Strategy Journal*.
- Lerner, J. (1994). The Syndication of Venture Capital Investments. FM. *The Journal of the Financial Management Association*, 23(3), 16–27.
- Lerner, J., & Nanda, R. (2020). Venture Capital's Role in Financing Innovation: What We Know and How Much We Still Need to Learn. *Journal of Economic Perspectives*, 34(3), 237–261.
- Lerner, J., Schoar, A., & Wongsunwai, W. (2007). Smart institutions, foolish choices: The limited performance puzzle. *The Journal of Finance*, 62(2), 731–764.
- Li, Y., Vertinsky, I. B., & Li, J. (2014). National distances, international experience, and venture capital investment performance. *Journal of Business Venturing*, 29(4), 471–489.
- Lindner, T., Puck, J., & Verbeke, A. (2020). Misconceptions about multicollinearity in international business research: Identification, consequences, and remedies. *Journal of International Business Studies*, 51(3), 283–298.
- Liu, Y., & Maula, M. (2016). Local partnering in foreign ventures: uncertainty, experiential learning, and syndication in cross-border venture capital investments. *Academy of Management Journal*, 59(4), 1407–1429.
- Lockett, A., & Wright, M. (1999). The syndication of private equity: Evidence from the UK. *Venture Capital*, 1(4), 303–324.
- Lockett, A., & Wright, M. (2001). The syndication of venture capital investments. *Omega-International Journal of Management Science*, 29(5), 375–390.
- Lu, Q., & Hwang, P. (2010). The impact of liability of foreignness on international venture capital firms in Singapore. *Asia Pacific Journal of Management*, 27(1), 81–97.
- Lutz, E., & Nörthemann, A. (2020). Venture Capital Syndication. In Douglas Cumming, & Hammer Benjamin (Eds.), *The Palgrave Encyclopedia of Private Equity*. Cham: Springer International Publishing.
- Madhavan, R., & Iriyama, A. (2009). Understanding global flows of venture capital: Human networks as the "carrier wave" of globalization. *Journal of International Business Studies*, 40(8), 1241–1259.
- Makela, M. M., & Maula, M. V. J. (2008). Attracting cross-border venture capital: the role of a local investor. *Entrepreneurship and Regional Development*, 20(3), 237–257.
- Makela, M. M., & Maula, M. V. J. (2006). Interorganizational commitment in syndicated cross-border venture capital investments. *Entrepreneurship Theory and Practice*, 30(2), 273–298.
- Manigart, S., Lockett, A., Meuleman, M., Wright, M., Landstrom, H., Bruining, H., Desbrières, P., & Hommel, U. (2006). Venture capitalists' decision to syndicate. *Entrepreneurship Theory and Practice*, 30(2), 131–153.
- Meuleman, M., Jaaskelainen, M., Maula, M. V. J., & Wright, M. (2017). Venturing into the unknown with strangers: Substitutes of relational embeddedness in cross-border partner selection in venture capital syndicates. *Journal of Business Venturing*, 32(2), 131–144.
- Meuleman, M., & Wright, M. (2011). Cross-border private equity syndication: Institutional context and learning. *Journal of Business Venturing*, 26(1), 35–48.
- Mindruta, D., Moeen, M., & Agarwal, R. (2016). A two-sided matching approach for partner selection and assessing complementarities in partners' attributes in inter-firm alliances. *Strategic Management Journal*, 37(1), 206–231.
- Mingo, S., Morales, F., & Dau, L. A. (2018). The interplay of national distances and regional networks: Private equity investments in emerging markets. *Journal of International Business Studies*, 49(3), 371–386.
- Moore, C. B., Payne, G. T., Bell, R. G., & Davis, J. L. (2015). Institutional Distance and Cross-Border Venture Capital Investment Flows. *Journal of Small Business Management*, 53(2), 482–500.
- Müllner, J., & Dorobantu, S. (2022). Overcoming political risk in developing economies through non-local debt. *Journal of International Business Policy*.
- Müller, J. & Puhr, H. 2021. Let me Google that for you: Capturing globalization using Google Trends. (<https://ssrn.com/abstract=3969013>).
- Park, S., & LiPuma, J. A. (2020). New venture internationalization: The role of venture capital types and reputation. *Journal of World Business*, 55(1), Article 101025.
- Pezeshkan, A., Smith, A., Fainshmidt, S., Zhang, J., & China. (2019). Risk, capabilities, and international venture capital syndication in. *International Journal of Entrepreneurial Behavior & Research*, 25(8), 1671–1684.
- Pichler, P., & Wilhelm, W. (2001). A Theory of the Syndicate: Form Follows Function. *The Journal of Finance*, 56(6), 2237–2264.
- Plagmann, C., & Lutz, E. (2019). Beggars or choosers? Lead venture capitalists and the impact of reputation on syndicate partner selection in international settings. *Journal of Banking Finance*, 100, 359–378.
- Preqin. 2021. Preqin Pro Database. (<https://www.preqin.com/>). 18 March 2021.
- Preqin. 2024. Preqin for academics. (<https://www.preqin.com/our-clients/academic-institutions/>) 17 June 2024.
- Puck, J., & Filatotchev, I. (2018). Finance and the multinational company: Building bridges between finance and global strategy research. *Global Strategy Journal*.
- Puck, J., Müllner, J., & Filatotchev, I. (2024). International finance. In *Encyclopedia of International Strategic Management* (pp. 211–217). Edward Elgar Publishing.
- Qiu, Z. Y., Chen, R., & Yang, Y. (2021). Going Abroad, Friends on Board: Cross-Border Venture Capital and Syndication Strategy in China. *Sage Open*, 11(2), 20.

- Schertler, A., & Tykvova, T. (2011). Venture capital and internationalization. *International Business Review*, 20(4), 423–439.
- Sethuram, S., Taussig, M., & Gaur, A. (2021). A multiple agency view of venture capital investment duration: The roles of institutions, foreignness, and alliances. *Global Strategy Journal*.
- Shenkar, O. (2001). Cultural Distance Revisited: Towards a More Rigorous Conceptualization and Measurement of Cultural Differences. *Journal of International Business Studies*, 32(3), 519–535.
- Sorenson, O., & Stuart, T. E. (2008). Bringing the context back in: Settings and the search for syndicate partners in venture capital investment networks. *Administrative Science Quarterly*, 53(2), 266–294.
- Sorenson, O., & Stuart, T. (2001). Syndication networks and the spatial distribution of venture capital investments. *American Journal of Sociology*, 106(6), 1546–1588.
- Stuart, T., & Sorenson, O. (2003a). The geography of opportunity: Spatial heterogeneity in founding rates and the performance of biotechnology firms. *Research Policy*, 32(2), 229–253.
- Stuart, T., & Sorenson, O. (2003b). Liquidity events and the geographic distribution of entrepreneurial activity. *Administrative Science Quarterly*, 48(2), 175–201.
- Tykvova, T., & Schertler, A. (2014). Does syndication with local venture capitalists moderate the effects of geographical and institutional distance? *Journal of International Management*, 20(4), 406–420.
- Wang, P. (2017). Syndication and foreignness: venture capital investments in emerging and developed markets. *Journal of International Management*, 23(1), 1–15.
- Wooldridge, Jeffrey M. (2015). "Control function methods in applied econometrics. *Journal of Human Resources*, 50(2), 420–445. DOI: 10.3368/jhr.50.2.420.
- Wright, M., Lockett, A., & Pruthi, S. (2002). Internationalization of western venture capitalists into emerging markets: Risk assessment and information in India. *Small Business Economics*, 19(1), 13–29.
- Wright, M., & Lockett, A. (2003). The structure and management of alliances: Syndication in the venture capital industry. *Journal of Management Studies*, 40(8), 2073–2102.
- Wright, M., Pruthi, S., & Lockett, A. (2005). International venture capital research: From cross-country comparisons to crossing borders. *International Journal of Management Reviews*, 7(3), 135–165.
- Young, S. (2003). International entrepreneurship research: What scope for international business theories? *Journal of International Entrepreneurship*, 1(1), 31–42.
- Zaheer, S. (1995). Overcoming the liability of foreignness. *Academy of Management Journal*, 38(2), 341–363.
- Zhang, J., & Pezeshkan, A. (2016). Host country network, industry experience, and international alliance formation: Evidence from the venture capital industry. *Journal of World Business*, 51(2), 264–277.
- Zhang, L., Gupta, A. K., & Hallen, B. L. (2017). The conditional importance of prior ties: A group-level analysis of venture capital syndication. *Academy of Management Journal*, 60(4), 1360–1386.
- Zhang, Y., Li, H., & Zhou, L. (2010). FDI spillovers in an emerging market: The role of foreign firms' country origin diversity and domestic firms' absorptive capacity. *Strategic Management Journal*, 31(9), 969–989.
- Zona, F., Gomez-Mejia, L. R., & Withers, M. C. (2018). Board Interlocks And Firm Performance: Toward a combined agency–resource dependence perspective. *Journal of Management*, 44(2), 589–618.